

Домашняя работа №2

№7.3

Дано:

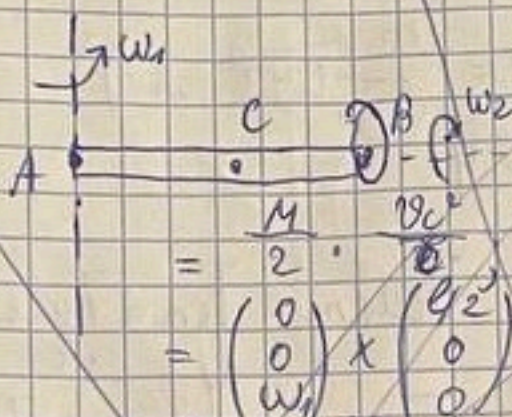
M, l

ω_1

r, m, ω_2

$T = ?$

Решение:



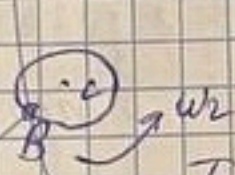
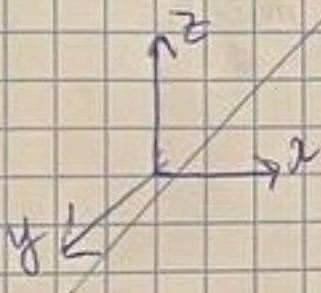
$$T_{cm} = \frac{M}{2} \cdot \frac{v_C^2}{2} = \frac{M \omega_1^2 l^2}{8}$$

$$T_{gucc} = \frac{m v_B^2}{2} + T_{omk}, \quad v_B^2 = \omega_1^2 l^2$$

$$T_{omk} = \frac{1}{2} \bar{\omega}^T J \bar{\omega} = \frac{1}{2} J_{AB} \omega_2^2 = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{m r^2}{2} \cdot \omega_2^2$$

$$T_{gucc} = \frac{m \omega_1^2 l^2}{2} + \frac{m r^2 \omega_2^2}{4}$$

$$T_{acm} = T_{cm} + T_{gucc} = \frac{M \omega_1^2 l^2}{8} + \frac{m \omega_1^2 l^2}{2} + \frac{m r^2 \omega_2^2}{4}$$



$$T_{acm} = T_{cm} + T_{gucc}$$

$$T_{cm} = \frac{m v_C^2}{2} + T_{omk} = \frac{1}{2} J_A \omega_1^2$$

$$= \frac{1}{2} \cdot \frac{1}{3} M l^2 \cdot \omega_1^2 = \frac{1}{6} M l^2 \omega_1^2$$

$$T_{gucc} = \frac{m v_C^2}{2} + T_{omk} = T_{omk} = \frac{1}{2} \bar{\omega}^T J_0 \bar{\omega} + \frac{m v_B^2}{2}$$

$$T_{omk} = \frac{1}{2} (A p^2 + B q^2 + C r^2)$$

$$A=0, B = \frac{m r^2}{2} + m l^2 + m l^2 \cos^2 \omega_2 t = \frac{3 m r^2}{2} + m l^2$$

$$C=0, B q^2 = \frac{3 m r^2}{2} + m l^2 + m l^2 \cos^2 \omega_2 t \quad B = \frac{3 m r^2}{2} + m (l^2 + r^2 \cos^2 \omega_2 t)$$

$$T_{omk} = \left(\frac{3 m r^2}{2} + m l^2 + m l^2 \cos^2 \omega_2 t \right) (\omega_1 + \omega_2)^2$$

$$q = \bar{\omega}_1 + \bar{\omega}_2 = \begin{pmatrix} 0 \\ \omega_2 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ \omega_1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ \omega_2 + \omega_1 \\ 0 \end{pmatrix}$$

$$T_{omk} = \frac{1}{2} B q^2 = \left(\frac{3 m r^2}{2} + m l^2 + m l^2 \cos^2 \omega_2 t \right) (\omega_1 + \omega_2)^2 =$$

$$= \frac{3 m r^2}{4} (\omega_1 + \omega_2)^2 + \frac{1}{2} m (l^2 + r^2 \cos^2 \omega_2 t + 2 l r \cos \omega_2 t) (\omega_1 + \omega_2)^2$$

$$= \frac{3}{4} m r^2 (\omega_1 + \omega_2)^2 + \frac{1}{2} m l^2 \omega_1^2 + m l r \cos \omega_2 t$$

$$T_{gucc} = \frac{m v_B^2}{2} + T_{omk} = \frac{m}{2} \omega_1^2 (l^2 + r^2 \cos^2 \omega_2 t) + T_{omk}$$

$$T = \frac{1}{2} (A p^2 + B q^2 + C r^2), \quad p=0, r=0, B = \frac{m r^2}{2} + m l^2 = \frac{3 m r^2}{2}, q = \omega_1$$

$$T_{\text{cm}} = \frac{1}{2} \cdot \frac{3mr^2}{2} \omega_2^2 = \frac{3}{4} mr^2 \omega_2^2$$

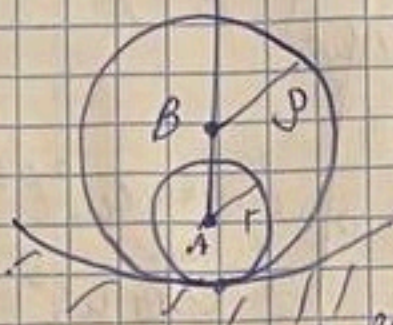
$$T_{\text{cm}} = \frac{1}{6} M l^2 \omega_1^2 + \frac{3}{4} mr^2 \omega_2^2 + \frac{m}{2} l^2 \omega_1^2 + m \omega_1^2 l r \cos \omega_1 t + \frac{m}{2} \omega_1^2 r^2 \cos^2 \omega_1 t$$

$$\frac{m}{2} v_c^2 = \frac{m}{2} (v_B + \omega_2 r)^2 = \frac{m}{2} (\omega_1 l + \omega_2 r)^2$$

N7.4

Дано:
 l, M
 ω, r, m
 $l = l + r$
 p, μ
 $T = ?$

Решение:



$$T_{\text{cm}} = T_{\text{cm}} + T_{\text{qu}} + T_{\text{op}}$$

$$T_{\text{cm}} = \frac{M v_c^2}{2} + T_{\text{cm}}$$

$$T_{\text{cm}} = \frac{1}{2} J_O \omega^2 = \frac{1}{2} \cdot \omega^2 \cdot \frac{1}{3} M l^2 = \frac{1}{6} M \omega^2 l^2$$

$$T_{\text{qu}} = \frac{m v_A^2}{2} + T_{\text{cm}} = \frac{m v_A^2}{2} + \frac{1}{6} M \omega^2 l^2$$

$$v_A = \omega l$$

$$T_{\text{qu}} = \frac{m \omega^2 l^2}{2} + \frac{1}{6} M \omega^2 l^2 = \frac{3}{4} m \omega^2 l^2$$

$$T_{\text{op}} = \frac{\mu v_B^2}{2} + T_{\text{cm}}$$

$$v_B = \omega (l + r - p) = \omega_0 \cdot p; \quad \omega_0 = \omega \cdot \frac{l + r - p}{p}$$

$$T_{\text{cm}} = \frac{1}{2} J_B \cdot \omega_0^2 = \frac{1}{2} \cdot \frac{1}{3} M p^2 \cdot \omega^2 \cdot \frac{(l + r - p)^2}{p^2} = \frac{1}{6} M \omega^2 (l + r - p)^2$$

$$T_{\text{op}} = \frac{\mu}{2} \omega^2 (l + r - p)^2 + \frac{1}{6} M \omega^2 (l + r - p)^2 = \frac{\mu}{2} \omega^2 (l + r - p)^2$$

$$T_c = \frac{1}{6} M \omega^2 l^2 + \frac{3}{4} m \omega^2 l^2 + \mu \omega^2 (l + r - p)^2$$

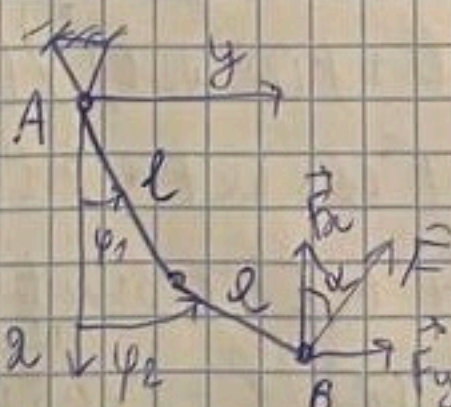
Ответ: $\frac{1}{6} M \omega^2 l^2 + \frac{3}{4} m \omega^2 l^2 + \mu \omega^2 (l + r - p)^2$

N7.59

Дано:

l, φ_1, φ_2
 $|F| = \text{const}$
 F - постоянная?

Решение:



Экстремум пометки и вычисления:

$$\frac{\partial F_2}{\partial \varphi_1} = \frac{\partial F_1}{\partial \varphi_2}$$

$$F_x = F \sin \alpha \cos \alpha$$

$$F_y = F \cos \alpha \sin \alpha$$

$$F_{\text{об}}^y = l \sin \varphi_1 + l \sin \varphi_2, \quad x_0 = l \cos \varphi_1 + l \cos \varphi_2$$

$$x_0 = l(\cos\varphi_1 + \cos\varphi_2)$$

$$y_0 = l(\sin\varphi_1 + \sin\varphi_2)$$

$$F_x = F \cdot \cos(\varphi_2 + \frac{\pi}{2}) = -F \sin\varphi_2$$

$$F_y = F \cdot \sin(\varphi_2 + \frac{\pi}{2}) = F \cos\varphi_2$$

$$\frac{\partial F_x}{\partial y} = \frac{\partial F_x}{\partial \varphi_1} \cdot \frac{\partial \varphi_1}{\partial y} + \frac{\partial F_x}{\partial \varphi_2} \cdot \frac{\partial \varphi_2}{\partial y} = -F \sin\varphi_2 \cdot \frac{\partial \varphi_2}{\partial y}$$

$$\frac{\partial F_y}{\partial x} = \frac{\partial F_y}{\partial \varphi_1} \cdot \frac{\partial \varphi_1}{\partial x} + \frac{\partial F_y}{\partial \varphi_2} \cdot \frac{\partial \varphi_2}{\partial x} = +F \cos\varphi_2 \cdot \frac{\partial \varphi_2}{\partial x}$$

$$\frac{\partial(x, y)}{\partial(\varphi_1, \varphi_2)} = \begin{pmatrix} -l\sin\varphi_1 & -l\sin\varphi_2 \\ l\cos\varphi_1 & l\cos\varphi_2 \end{pmatrix}$$

$$\frac{\partial(\varphi_1, \varphi_2)}{\partial(x, y)} = \begin{pmatrix} l\cos\varphi_2 & -l\cos\varphi_1 \\ l\sin\varphi_2 & -l\sin\varphi_1 \end{pmatrix}^T$$

$$\cdot \frac{1}{l^2(\sin\varphi_1 \cos\varphi_2 - \cos\varphi_1 \sin\varphi_2)} = \begin{pmatrix} -\frac{\sin\varphi_2}{\sin\varphi_1 \cos\varphi_2 - \cos\varphi_1 \sin\varphi_2} & \frac{\sin\varphi_1}{\sin\varphi_1 \cos\varphi_2 - \cos\varphi_1 \sin\varphi_2} \\ -\cos\varphi_1 & \cos\varphi_2 \end{pmatrix}$$

$$\varphi_1 \begin{pmatrix} \cos\varphi_2 & \sin\varphi_2 \\ -\sin\varphi_1 & -\cos\varphi_1 \end{pmatrix} \cdot \frac{1}{l(\sin\varphi_2 \cos\varphi_1 - \sin\varphi_1 \cos\varphi_2)}$$

$$\frac{\partial \varphi_2}{\partial x} = \frac{\sin\varphi_2 - \cos\varphi_1}{l(\sin\varphi_2 \cos\varphi_1 - \sin\varphi_1 \cos\varphi_2)}, \quad \frac{\partial \varphi_2}{\partial y} = \frac{\cos\varphi_2 - \sin\varphi_1}{l(\sin\varphi_2 \cos\varphi_1 - \sin\varphi_1 \cos\varphi_2)}$$

$$\frac{\partial F_x}{\partial y} = \frac{F \cos\varphi_2 \sin\varphi_2 \cos\varphi_1}{l(\sin\varphi_2 \cos\varphi_1 - \sin\varphi_1 \cos\varphi_2)}; \quad \frac{\partial F_y}{\partial x} = \frac{F \cos\varphi_2 \sin\varphi_2 \cos\varphi_1}{l(\sin\varphi_2 \cos\varphi_1 - \sin\varphi_1 \cos\varphi_2)}$$

$$\frac{\partial F_x}{\partial y} \neq \frac{\partial F_y}{\partial x} \Rightarrow \text{не потенциальное.}$$

Ответ: не потенциальное.

Дано:

$$m, 2L$$

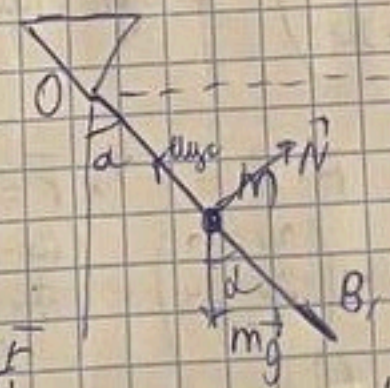
$$\omega$$

$$\omega = \text{const}$$

$$t = ?$$

$$\vec{v} = \vec{v}_0 + \vec{\omega} \times \vec{r}$$

Решение:



$$A \quad J_0 = \frac{1}{3} m l^2 = \frac{4}{3} m L^2$$

$$\vec{M}_0 = m\vec{g} \times \vec{r} = m\vec{g} \times \vec{r} + J_0 \vec{\omega}$$

$$K_0 = \vec{r} \times m\vec{v} = J_0 \cdot \vec{\omega} = (mr^2 + J_0) \vec{\omega}$$

$$K_0 = \vec{r} \times m\vec{v} = J_0 \vec{\omega} = mr^2 \vec{\omega} + J_0 \vec{\omega} = (mr^2 + J_0) \vec{\omega}$$

$$\frac{dK_0}{dt} = M_0; \quad (mr^2 + J_0) \left(\frac{d\omega}{dt} \vec{e} + 0 \right) = m\vec{r} \times m\vec{g} + J_0 \vec{\omega}$$

$$mr^2 \vec{e} + J_0 \vec{e} = \vec{r} \times m\vec{g} + J_0 \vec{e}; \quad mr^2 \vec{e} = \vec{r} \times m\vec{g}; \quad \vec{e} = \vec{r} \times \vec{g}$$

$$\vec{e} = \frac{1}{r^2} [\vec{r} \times \vec{g}]; \quad \vec{\omega} = \frac{1}{r^2} [\vec{r} \times \vec{g}]$$

$$2r\vec{r} \cdot \vec{\omega} = \vec{r} \times \vec{g}; \quad 2r^2 \omega = g \sin \alpha = g \sin \alpha \omega$$

$$K_{Ca} = K_{Aa}, K_{Cy} = K_{Ay}, K_{Cz} = K_{Az}$$

N 11.24

Дано:

m, a

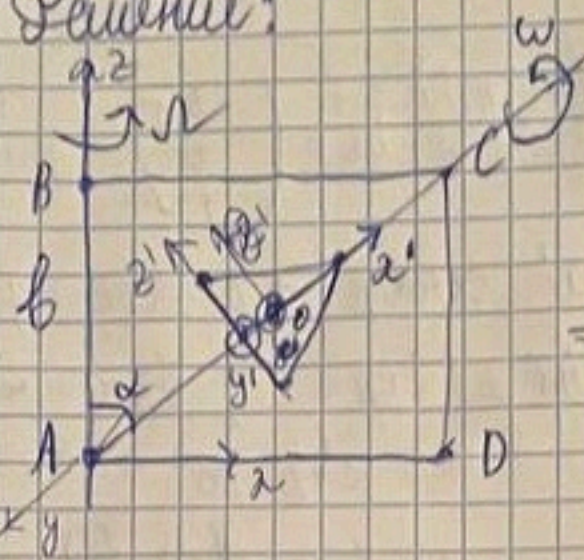
ω, Ω

$T = ?$

$AB = b$

$\angle BAC = \alpha$
 $AO = OC$

Решение:



$$T_{\text{м}} = \frac{m v_0^2}{2} + T_{\text{отн}}$$

$$\vec{v}_0 = \vec{v}_A + \vec{\omega} \times \vec{AO} = \begin{pmatrix} 0 \\ 0 \\ \Omega \end{pmatrix} \times \begin{pmatrix} a \cos \alpha/2 \\ a \sin \alpha/2 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ \Omega a \sin \alpha/2 \\ 0 \end{pmatrix}$$

$$|v_0| = \Omega a \sin \frac{\alpha}{2}$$

$$T_{\text{отн}} = (A p^2 + B q^2 + C r^2) \cdot \frac{1}{2}$$

$$A = \int r^2 dm = \int_0^a \int_0^{\sqrt{3}/2 a} \int_0^{2a - \sqrt{3}z} \frac{1}{2} (2a - \sqrt{3}z)^2 \rho dz dx dy =$$

$$= \frac{1}{2} \rho \int_0^a \int_0^{\sqrt{3}/2 a} (2a - \sqrt{3}z)^2 dz dy = \frac{1}{2} \rho \int_0^a \left(\frac{3}{2} a^2 - \sqrt{3} a z + \frac{3}{2} z^2 \right) dy dy =$$

$$= \frac{1}{2} \rho \cdot \left(\frac{3}{2} a^2 y - \sqrt{3} a z y + \frac{3}{2} z^2 y \right) \Big|_0^{\sqrt{3}/2 a} = \frac{1}{2} \rho \cdot \left(\frac{3}{2} a^2 \frac{\sqrt{3}}{2} a - \sqrt{3} a z \frac{\sqrt{3}}{2} a + \frac{3}{2} z^2 \frac{\sqrt{3}}{2} a \right) =$$

$$= \frac{1}{2} \rho \cdot \left(\frac{3}{2} a^3 \frac{\sqrt{3}}{2} - \frac{3}{2} a^3 \frac{\sqrt{3}}{2} + \frac{3}{2} a^3 \frac{\sqrt{3}}{2} \right) = \frac{1}{2} \rho \cdot \frac{3}{2} a^3 \frac{\sqrt{3}}{2} = \frac{3}{8} \rho a^3 \sqrt{3}$$

$$= \frac{3}{8} \rho a^3 \sqrt{3} = \frac{3}{8} m a^2 \sqrt{3}$$

$$A_2 = \int r^2 dm = \int_0^a \int_0^{\sqrt{3}/2 a} \int_0^{2a - \sqrt{3}z} \frac{1}{2} (2a - \sqrt{3}z)^2 \rho dz dx dy =$$

$$= \frac{1}{2} \rho \int_0^a \int_0^{\sqrt{3}/2 a} (2a - \sqrt{3}z)^2 dz dy = \frac{1}{2} \rho \int_0^a \left(\frac{3}{2} a^2 - \sqrt{3} a z + \frac{3}{2} z^2 \right) dy dy =$$

$$= \frac{1}{2} \rho \cdot \left(\frac{3}{2} a^2 y - \sqrt{3} a z y + \frac{3}{2} z^2 y \right) \Big|_0^{\sqrt{3}/2 a} = \frac{1}{2} \rho \cdot \left(\frac{3}{2} a^2 \frac{\sqrt{3}}{2} a - \sqrt{3} a z \frac{\sqrt{3}}{2} a + \frac{3}{2} z^2 \frac{\sqrt{3}}{2} a \right) =$$

$$= \frac{1}{2} \rho \cdot \left(\frac{3}{2} a^3 \frac{\sqrt{3}}{2} - \frac{3}{2} a^3 \frac{\sqrt{3}}{2} + \frac{3}{2} a^3 \frac{\sqrt{3}}{2} \right) = \frac{1}{2} \rho \cdot \frac{3}{2} a^3 \frac{\sqrt{3}}{2} = \frac{3}{8} \rho a^3 \sqrt{3}$$

$$= \frac{3}{8} \rho a^3 \sqrt{3} = \frac{3}{8} m a^2 \sqrt{3}$$

$$= \frac{3}{8} m a^2 \sqrt{3}$$

$$I_y = I_2 + I_1 = \left(\frac{1}{8} + \frac{3}{8} \right) m a^2 = \frac{1}{2} m a^2$$

$$\vec{\omega}_{\text{отн}} = \vec{\omega} + \vec{\Omega} = \begin{pmatrix} \omega + \Omega \cos \alpha \\ \Omega \sin \alpha \sin \omega t \\ \Omega \sin \alpha \cos \omega t \end{pmatrix}$$

$$T_{\text{mean}} = \frac{1}{2} m a^2 \left((\omega^2 + \Omega \cos \alpha) \frac{2\pi}{8} + \Omega^2 \sin^2 \alpha \cos^2 \omega t \frac{\pi}{8} + \frac{1}{16} \Omega^2 \sin^2 \alpha \sin^2 \omega t \right) \\ = \frac{1}{16} m a^2 \left(\omega^2 + 2\Omega \omega \cos \alpha + \Omega^2 \cos^2 \alpha + \Omega^2 \sin^2 \alpha + \Omega^2 \sin^2 \alpha \sin^2 \omega t \right) =$$

$$T = \frac{m \Omega^2 b^2}{8} \tan^2 \alpha + \frac{1}{16} m a^2 (\omega^2 + 2\Omega \omega \cos \alpha + 2\Omega^2 \sin^2 \alpha + \Omega^2 (1 + \sin^2 \alpha \sin^2 \omega t))$$

$$T_{\text{tot}} = \frac{1}{2} m \dot{\alpha}^2 \left((\omega^2 + \Omega \cos \alpha) \frac{1}{8} + \Omega^2 \sin^2 \alpha \cos^2 \omega t \frac{9}{8} + \frac{1}{16} \Omega^2 \sin^2 \alpha \sin^2 \omega t \right) =$$

$$= \frac{1}{16} m \dot{\alpha}^2 (\omega^2 + 2\Omega \omega \cos \alpha + \Omega^2 \cos^2 \alpha + \Omega^2 \sin^2 \alpha + \Omega^2 \sin^2 \alpha \sin^2 \omega t) =$$

$$T = \frac{m \Omega^2 b^2}{8} \epsilon g^2 \alpha + \frac{1}{16} m \dot{\alpha}^2 (\omega^2 + 2\Omega \omega \cos \alpha + 2\Omega^2 \sin^2 \alpha + \Omega^2 (1 + \sin^2 \alpha \sin^2 \omega t))$$

N 11.74

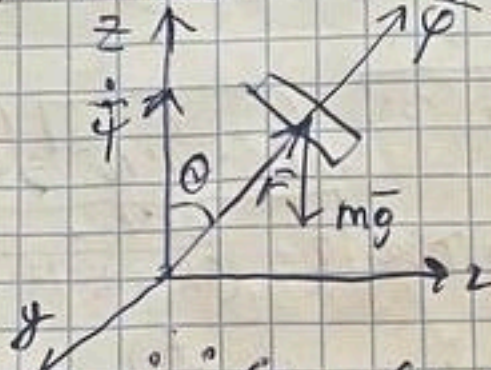
Дано:

r, θ

$\dot{\psi}$

$\dot{\psi} - ?$

Решение:



$$A = B \neq C$$

$$\bar{M} = \dot{\psi} \times \dot{\psi} [C + (C-A) \frac{\dot{\psi}}{\dot{\psi}} \cos \theta]$$

$$\bar{M} = \text{mg} \bar{r} \times \text{mg}$$

$$\bar{r} \parallel \dot{\psi}, \quad \dot{\psi} \parallel -\text{mg}$$

$$\dot{\psi} \dot{\psi} (C + (C-A) \frac{\dot{\psi}}{\dot{\psi}} \cos \theta) = +m g r$$

$$C \dot{\psi} \dot{\psi} + (C-A) \cos \theta \dot{\psi}^2 = +m g r$$

$$\dot{\psi} = \frac{+m g r + (A-C) \cos \theta \dot{\psi}^2}{C \dot{\psi}}$$

$$C = \frac{1}{2} m r^2, \quad A = B = \frac{1}{4} m r^2$$

$$\dot{\psi} = \frac{m g r - \frac{1}{4} m r^2 \cos \theta \dot{\psi}^2}{\frac{1}{2} m r^2 \dot{\psi}} = \frac{2g}{r \dot{\psi}} - \frac{\cos \theta \dot{\psi}}{2}$$

Ответ: $\dot{\psi} = \frac{2g}{r \dot{\psi}} - \frac{\cos \theta \dot{\psi}}{2}$

N 11.108

Дано:

$$A = B \neq C$$

p_0, q_0, r_0

$\alpha - ?$

$\dot{\psi}, \dot{\psi} - ?$

Решение:

Случай Эйлера:

$$C \bar{r}_0 = (K, e_3); \quad C r_0 = |K| \cos \theta$$

$$\cos \theta = \frac{C r_0}{|K|}$$

$$\bar{K} = J \bar{\omega}; \quad \bar{K} = \begin{pmatrix} A & 0 & 0 \\ 0 & B & 0 \\ 0 & 0 & C \end{pmatrix} \begin{pmatrix} p_0 \\ q_0 \\ r_0 \end{pmatrix} = \begin{pmatrix} A p_0 & 0 & 0 \\ 0 & B q_0 & 0 \\ 0 & 0 & C r_0 \end{pmatrix} \begin{pmatrix} A p_0 \\ B q_0 \\ C r_0 \end{pmatrix}$$

$$|K| = \sqrt{A^2 p_0^2 + B^2 q_0^2 + C^2 r_0^2}$$

$$\theta = \arccos \left(\frac{C r_0}{\sqrt{A^2 p_0^2 + B^2 q_0^2 + C^2 r_0^2}} \right)$$

$$\bar{K} = A \bar{p}_0 + B \bar{q}_0 + C \bar{r}_0$$

$$\dot{\psi} = \frac{|K|}{A} = \sqrt{p_0^2 + \frac{B^2}{A^2} q_0^2 + \frac{C^2}{A^2} r_0^2}; \quad \dot{\psi} = \frac{|K|}{A}$$

$$|K| \cos \psi = C(\dot{\psi} + \dot{\psi} \cos \theta); \quad C r_0 = C(\dot{\psi} + \dot{\psi} \cos \theta); \quad r_0 = \dot{\psi} + \dot{\psi} \cos \theta$$

Ответ: $\theta = \arccos \left(\frac{C r_0}{\sqrt{A^2 p_0^2 + B^2 q_0^2 + C^2 r_0^2}} \right)$

$$\dot{\psi} = \sqrt{p_0^2 + \frac{B^2}{A^2} q_0^2 + \frac{C^2}{A^2} r_0^2}$$

$$\dot{\psi} = r_0 \left(1 - \frac{C}{A} \right)$$

$A = B$

$$\dot{\psi} = r_0 - \dot{\psi} \cos \theta = \frac{|K|}{A} \frac{C r_0}{|K|} = r_0 \left(1 - \frac{C}{A} \right)$$

Dano:

h, R

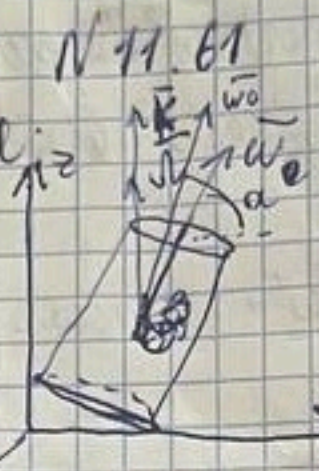
ω_0, d

$\dot{\psi} = ?$

$\dot{\psi} = ?$

$\theta = ?$

Pemenuk:



$$\vec{\omega}_0 = \vec{\omega} + \vec{\pi}$$

Some maximum and

Chosogn. pergh. mpegic.

$$\cos \theta = \frac{C \vec{\omega}_0}{|\vec{K}|} = \frac{C \vec{\omega}_0}{\sqrt{A \vec{\omega}_0^2 + C \vec{\omega}_0^2}} = \frac{C \vec{\omega}_0}{\sqrt{A \vec{\omega}_0^2 + C \vec{\omega}_0^2}}$$

$$\tan \theta = \frac{A \sqrt{A \vec{\omega}_0^2 + C \vec{\omega}_0^2}}{C \vec{\omega}_0} = \frac{A \sqrt{A \vec{\omega}_0^2 + C \vec{\omega}_0^2}}{C \vec{\omega}_0}$$

$$\dot{\psi} = \frac{|\vec{K}|}{A} = \frac{\sqrt{A \vec{\omega}_0^2 + C \vec{\omega}_0^2}}{A} = \sqrt{\vec{\omega}_0^2 \cos^2 \theta + \frac{1}{16} m^2 \vec{\omega}_0^4 + \frac{1}{4} m^2 \vec{\omega}_0^2 + \frac{1}{16} m^2 \vec{\omega}_0^4}$$

$$K = \frac{1}{2} \sqrt{A \vec{\omega}_0^2 + C \vec{\omega}_0^2}, \quad \vec{p}_0 = \vec{\omega}_0 \cos \theta, \quad \dot{\psi} = \vec{r}_0 \left(1 - \frac{A}{C} \right) = \omega_0 \sin \theta \left(1 - \frac{A}{C} \right)$$

$$A = \frac{1}{4} m R^2, \quad C = \frac{1}{2} m R^2$$

$$\dot{\psi} = \omega_0 \sin \theta \left(1 - \frac{A}{C} \right) = \omega_0 \sin \theta \left(1 - \frac{1/4 m R^2}{1/2 m R^2} \right) = \omega_0 \sin \theta \left(1 - \frac{1}{2} \right) = \frac{1}{2} \omega_0 \sin \theta$$

Dano:

R, m

$C = \frac{1}{2} m R^2$

$\psi_0 = 0 = \psi_0, \quad \theta_0 = \frac{\pi}{6}$

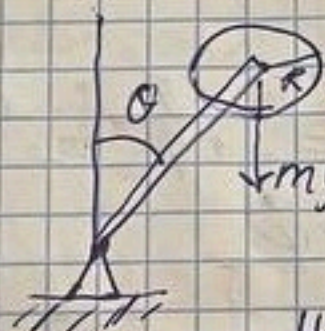
$\dot{\psi}_0 = 0, \quad \dot{\theta}_0 = 4 \sqrt{\frac{g}{R}}$

$\dot{\psi}_0 = 0, \quad \theta_{\max} = ?$

$\dot{\psi}(\theta) = ?$

$\dot{\psi}(\theta) = ?$

Pemenuk:



$$\vec{M} = \vec{r} \times \vec{p} = (c + (c - A) \frac{\dot{\psi}}{\dot{\theta}} \cos \theta) m \vec{v}$$

$$\vec{M} = \vec{r} \times m \vec{g} = \dot{\psi} \dot{\theta} (c + (c - A) \frac{\dot{\psi}}{\dot{\theta}} \cos \theta) = m g l$$

$$\dot{\psi} \dot{\theta} c + \dot{\psi}^2 (c - A) \cos \theta = m g l$$

$$C = \frac{1}{2} m R^2, \quad A = \frac{1}{4} m R^2$$

$$\frac{1}{4} \dot{\psi}^2 m R^2 + \frac{1}{4} \dot{\psi}^2 m R^2 \cos \theta = m g l$$

$$\frac{1}{4} R^2 (2 \dot{\psi} \dot{\theta} + \dot{\psi}^2 \cos \theta) = g l$$

$$2 \dot{\psi} \dot{\theta} + \dot{\psi}^2 \cos \theta = \frac{4 g l}{R^2} = \frac{4 g \sqrt{3} R}{2 R^2} = \frac{2 \sqrt{3} g}{R}$$

$$E = T + \Pi = \text{const}$$

$$T = \frac{1}{2} A \dot{\rho}^2 + \frac{1}{2} C \dot{\theta}^2$$

$$\Pi = m g l \cos \theta$$

$$\rho_0 = 0$$

$$\theta_0 = 0$$

$$E = \frac{1}{2} \cdot \frac{1}{2} m R^2 \cdot 16 \frac{g}{R} + m g l \frac{\sqrt{3}}{2} = \frac{4 m g R}{3} + \frac{1}{2} m g R = \frac{10}{3} m g R$$

$$= 4 m g R + \frac{1}{2} m g R = \frac{9}{2} m g R = \text{maksimum } \theta \text{ dan minimum}$$

$$E = \frac{1}{2} \cdot \frac{1}{4} m R^2 (\dot{\theta}^2 + \dot{\psi}^2 \sin^2 \theta) + \frac{1}{2} \cdot \frac{1}{2} m R^2 (\dot{\varphi}^2 + \dot{\psi}^2 \cos^2 \theta + 2 \dot{\varphi} \dot{\psi} \cos \theta) + m g \frac{3}{4} m g R = \frac{1}{8} m R^2 \dot{\theta}^2 + \frac{1}{4} m R^2 \dot{\psi}^2 + \frac{1}{8} m R^2 \dot{\varphi}^2 + \frac{1}{2} m R^2 \dot{\varphi} \dot{\psi} \cos \theta + \frac{3}{4} m g R$$

$$\frac{1}{8} R^2 \dot{\theta}^2 + \frac{1}{4} \dot{\psi}^2 + \frac{1}{8} \dot{\varphi}^2 \sin^2 \theta + \frac{1}{4} R^2 \dot{\varphi} + \frac{1}{2} \dot{\varphi} \dot{\psi} \cos \theta = 4 g R$$

$$R \left(\frac{1}{8} \dot{\theta}^2 + \frac{1}{4} \dot{\psi}^2 + \frac{1}{8} \dot{\varphi}^2 \sin^2 \theta + \frac{1}{4} \dot{\varphi} + \frac{1}{2} \dot{\varphi} \dot{\psi} \cos \theta \right) = \frac{4 g}{R}$$

$$\dot{\varphi} = \frac{\frac{4 g}{R} - \dot{\psi}^2 \cos \theta}{2 \dot{\psi}}$$

N/2

L, M, N

$\frac{d}{dt}$ - производная
 $\bar{}$ - вектор

$$L \circ (M \circ N) = (L \circ M) \circ N \text{ гок-мб}$$

$$L = \bar{x}_0 \bar{x}_0 + \bar{x}_1 \bar{x}_1 + \bar{x}_2 \bar{x}_2 + \bar{x}_3 \bar{x}_3 = \bar{x}_0 + \bar{x}$$

$$M = \bar{x}_1 \bar{x}_1 + \bar{x}_1 \bar{x}_2, N = \bar{x}_2 + \bar{x}_2$$

$$1) L \circ (M \circ N) = L \circ (\bar{x}_1 \bar{x}_2 - (\bar{x}_1 \bar{x}_2) + \bar{x}_1 \bar{x}_2 + \bar{x}_2 \bar{x}_1 - [\bar{x}_1, \bar{x}_2]) = \\ = \bar{x}_0 (\bar{x}_1 \bar{x}_2 - (\bar{x}_1 \bar{x}_2)) + (\bar{x}_0, \bar{x}_1 \bar{x}_2 + \bar{x}_2 \bar{x}_1 - [\bar{x}_1, \bar{x}_2]) + \\ + \bar{x}_0 (\bar{x}_1 \bar{x}_2 + \bar{x}_2 \bar{x}_1 - [\bar{x}_1, \bar{x}_2]) + (\bar{x}_1 \bar{x}_2 - (\bar{x}_1 \bar{x}_2)) \bar{x}_0 - [\bar{x}_0, \bar{x}_1 \bar{x}_2 + \\ + \bar{x}_2 \bar{x}_1 - [\bar{x}_1, \bar{x}_2]] = \bar{x}_0$$

$$2) (L \circ M) \circ N = (\bar{x}_0 \bar{x}_1 + \bar{x}_0 \bar{x}_2 - [\bar{x}_0, \bar{x}_1]) \circ (\bar{x}_2 + \bar{x}_2) =$$

$$= \bar{x}_0 \bar{x}_1 \bar{x}_2 - (\bar{x}_0, \bar{x}_1) \cdot \bar{x}_2 = (\bar{x}_0 \bar{x}_1, \bar{x}_2 \bar{x}_0 - [\bar{x}_0, \bar{x}_1], \bar{x}_2) +$$

$$+ \bar{x}_0 \bar{x}_2 (\bar{x}_0 \bar{x}_1 - (\bar{x}_0, \bar{x}_1)) + (\bar{x}_0 \bar{x}_1 + \bar{x}_0 \bar{x}_2 - [\bar{x}_0, \bar{x}_1]) \bar{x}_2 =$$

$$- [\bar{x}_0 \bar{x}_1 + \bar{x}_0 \bar{x}_2 - [\bar{x}_0, \bar{x}_1], \bar{x}_2] =$$

$$1) \text{ Крайнее значение: } \bar{x}_0 (\bar{x}_1 \bar{x}_0 \bar{x}_2 - \bar{x}_0 (\bar{x}_1, \bar{x}_2) - (\bar{x}_0, \bar{x}_1 \bar{x}_2 + \bar{x}_2 \bar{x}_1 - [\bar{x}_1, \bar{x}_2])) \\ \bar{x}_0 \bar{x}_1 \bar{x}_2 - \bar{x}_2 (\bar{x}_0, \bar{x}_1) - (\bar{x}_2, \bar{x}_0 \bar{x}_1 + \bar{x}_1 \bar{x}_0 - [\bar{x}_0, \bar{x}_1])$$

2) Крайнее значение:

$$(\bar{x}_0, \bar{x}_1 \bar{x}_2 + \bar{x}_2 \bar{x}_1 - [\bar{x}_1, \bar{x}_2]) = \bar{x}_1 (\bar{x}_0, \bar{x}_2) + \bar{x}_2 (\bar{x}_0, \bar{x}_1) - (\bar{x}_0, \bar{x}_1, \bar{x}_2)$$

$$(\bar{x}_2, \bar{x}_0 \bar{x}_1 + \bar{x}_1 \bar{x}_0 - [\bar{x}_0, \bar{x}_1]) = \bar{x}_0 (\bar{x}_1, \bar{x}_2) + \bar{x}_1 (\bar{x}_0, \bar{x}_2) - (\bar{x}_2, \bar{x}_0, \bar{x}_1)$$

Крайние значения равны нулю ✓, проверили векторное

1) Вектор. часть: $\vec{r}_0'(\vec{r}_1\vec{r}_2 + \vec{r}_2\vec{r}_1 - [\vec{r}_1, \vec{r}_2]) + (\vec{r}_1\vec{r}_2 - (\vec{r}_1, \vec{r}_2))\vec{r}_0 - [\vec{r}_0, \vec{r}_1\vec{r}_2 + \vec{r}_2\vec{r}_1 - [\vec{r}_1, \vec{r}_2]]$

2) Вектор. часть: $\vec{r}_2(\vec{r}_0\vec{r}_1 - (\vec{r}_0, \vec{r}_1)) + \vec{r}_1(\vec{r}_0\vec{r}_2 + \vec{r}_0\vec{r}_1 - [\vec{r}_0, \vec{r}_2]) - [\vec{r}_0\vec{r}_1 + \vec{r}_1\vec{r}_0 - [\vec{r}_0, \vec{r}_1], \vec{r}_2]$

1) $\vec{r}_0'\vec{r}_1\vec{r}_2 + \vec{r}_0'\vec{r}_2\vec{r}_1 + \vec{r}_1'\vec{r}_2\vec{r}_0 - \vec{r}_0'[\vec{r}_1, \vec{r}_2] - (\vec{r}_0, \vec{r}_1, \vec{r}_2) - \vec{r}_1'[\vec{r}_0, \vec{r}_2] - \vec{r}_2'[\vec{r}_0, \vec{r}_1] + (\vec{r}_0, \vec{r}_1, \vec{r}_2)$

2) $\vec{r}_0'\vec{r}_1\vec{r}_2 - (\vec{r}_0, \vec{r}_1, \vec{r}_2) + \vec{r}_0'\vec{r}_2\vec{r}_1 + \vec{r}_1'\vec{r}_2\vec{r}_0 - \vec{r}_2'[\vec{r}_0, \vec{r}_1] - \vec{r}_0'[\vec{r}_1, \vec{r}_2] - \vec{r}_1'[\vec{r}_0, \vec{r}_2] + (\vec{r}_0, \vec{r}_1, \vec{r}_2)$

Векторные части также равны $\Rightarrow$ ассоциат. доказана
N 4.66

Дано:	Решение:
φ, θ, ψ	1) $\nearrow Oz$ на φ , 2) $\nearrow Ox'$ на θ , 3) $\nearrow Oz''$ на ψ
$\vec{r}_1, \vec{r}_2, \vec{r}_3$ - ?	$\vec{L} = \cos \frac{\varphi}{2} + \vec{e} \sin \frac{\varphi}{2}$
$\vec{r}_0, \vec{r}_1, \vec{r}_2, \vec{r}_3$ - ?	$\vec{L}_1 = \cos \frac{\varphi}{2} + \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \sin \frac{\varphi}{2}$
	$\vec{L}_2 = \cos \frac{\theta}{2} + \begin{pmatrix} \cos \varphi \\ \sin \varphi \\ 0 \end{pmatrix} \sin \frac{\theta}{2}$
	$\vec{L}_3 = \cos \frac{\psi}{2} + \begin{pmatrix} \sin \varphi \sin \theta \\ \cos \varphi \sin \theta \\ \cos \theta \end{pmatrix} \sin \frac{\psi}{2}$

$\vec{L} = \vec{L}_3 \circ \vec{L}_2 \circ \vec{L}_1 =$

$$\begin{pmatrix} \cos \frac{\varphi}{2} \\ 0 \\ 0 \\ \sin \frac{\varphi}{2} \end{pmatrix} \circ \begin{pmatrix} \cos \frac{\theta}{2} \\ \cos \varphi \sin \frac{\theta}{2} \\ \sin \varphi \sin \frac{\theta}{2} \\ 0 \end{pmatrix} \circ \begin{pmatrix} \cos \frac{\psi}{2} \\ \sin \varphi \sin \theta \sin \frac{\psi}{2} \\ \cos \varphi \sin \theta \sin \frac{\psi}{2} \\ \cos \theta \sin \frac{\psi}{2} \end{pmatrix} \circ \begin{pmatrix} \cos \frac{\psi}{2} \\ \cos \varphi \sin \frac{\psi}{2} \\ \sin \varphi \sin \frac{\psi}{2} \\ 0 \end{pmatrix} \circ \begin{pmatrix} \cos \frac{\psi}{2} \\ 0 \\ 0 \\ \sin \frac{\psi}{2} \end{pmatrix} =$$

$$= \begin{pmatrix} \cos \frac{\varphi}{2} \\ \sin \varphi \sin \theta \sin \frac{\psi}{2} \\ \cos \varphi \sin \theta \sin \frac{\psi}{2} \\ \cos \theta \sin \frac{\psi}{2} \end{pmatrix} \circ \begin{pmatrix} \cos \frac{\theta}{2} \cos \frac{\varphi}{2} = 0 \\ 0 + \cos \varphi \sin \frac{\theta}{2} \cos \frac{\varphi}{2} = \sin \frac{\varphi}{2} \sin \varphi \sin \frac{\theta}{2} \\ 0 + \cos \varphi \sin \frac{\theta}{2} \sin \varphi \sin \frac{\varphi}{2} + \sin \frac{\varphi}{2} \cos \varphi \sin \frac{\theta}{2} \\ \cos \frac{\theta}{2} \sin \frac{\varphi}{2} + 0 - 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} \circ \begin{pmatrix} \cos \frac{\psi}{2} \sin \frac{\varphi}{2} \\ \sin \frac{\theta}{2} \cos(1,54) \\ \sin \frac{\theta}{2} \sin(1,54) \\ \cos \frac{\theta}{2} \sin \frac{\varphi}{2} \end{pmatrix} =$$

$\Rightarrow \begin{pmatrix} \cos \frac{\varphi}{2} \cos \frac{\theta}{2} \sin \frac{\psi}{2} - \end{pmatrix}$

активная точка
зрения неудобна
воспроизведения
пассивной

$$\Lambda_1 = \cos \frac{\varphi}{2} + \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \sin \frac{\varphi}{2}, \quad \Lambda_2 = \cos \frac{\vartheta}{2} + e_2 \sin \frac{\vartheta}{2} = \cos \frac{\vartheta}{2} + \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \sin \frac{\vartheta}{2}$$

$$\Lambda_3 = \cos \frac{\varphi}{2} + e_2 \sin \frac{\varphi}{2} = \cos \frac{\varphi}{2} + \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \sin \frac{\varphi}{2}$$

$\Lambda_3 \circ \Lambda_2 \circ \Lambda_1 = \Lambda_{\text{рез}}$ в упр.; $\Lambda_{\text{рез}} = \Lambda_1 \circ \Lambda_2 \circ \Lambda_3$ - в расчб.

$$\Lambda_{\text{рез}} = \begin{pmatrix} \cos \frac{\varphi}{2} & 0 & 0 \\ 0 & \cos \frac{\vartheta}{2} & 0 \\ 0 & 0 & \sin \frac{\varphi}{2} \end{pmatrix} \circ \begin{pmatrix} \cos \frac{\vartheta}{2} & \sin \frac{\vartheta}{2} & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \sin \frac{\varphi}{2} \end{pmatrix} = \begin{pmatrix} \cos \frac{\varphi}{2} \cos \frac{\vartheta}{2} & \cos \frac{\varphi}{2} \sin \frac{\vartheta}{2} & 0 \\ \cos \frac{\varphi}{2} \sin \frac{\vartheta}{2} & \sin \frac{\varphi}{2} \cos \frac{\vartheta}{2} & 0 \\ -\sin \frac{\varphi}{2} \cos \frac{\vartheta}{2} & -\sin \frac{\varphi}{2} \sin \frac{\vartheta}{2} & \sin \frac{\varphi}{2} \end{pmatrix} \circ \begin{pmatrix} \cos \frac{\varphi}{2} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \sin \frac{\varphi}{2} \end{pmatrix} = \begin{pmatrix} \cos \frac{\varphi}{2} \cos \frac{\vartheta}{2} \cos \frac{\varphi}{2} - \sin \frac{\varphi}{2} \sin \frac{\vartheta}{2} \cos \frac{\varphi}{2} + \cos \frac{\varphi}{2} \sin \frac{\vartheta}{2} \sin \frac{\varphi}{2} + \sin \frac{\varphi}{2} \cos \frac{\vartheta}{2} \sin \frac{\varphi}{2} & \cos \frac{\varphi}{2} \cos \frac{\vartheta}{2} \sin \frac{\varphi}{2} - \sin \frac{\varphi}{2} \sin \frac{\vartheta}{2} \sin \frac{\varphi}{2} + \cos \frac{\varphi}{2} \sin \frac{\vartheta}{2} \cos \frac{\varphi}{2} + \sin \frac{\varphi}{2} \cos \frac{\vartheta}{2} \cos \frac{\varphi}{2} & \cos \frac{\varphi}{2} \cos \frac{\vartheta}{2} \sin \frac{\varphi}{2} - \sin \frac{\varphi}{2} \sin \frac{\vartheta}{2} \sin \frac{\varphi}{2} + \cos \frac{\varphi}{2} \sin \frac{\vartheta}{2} \cos \frac{\varphi}{2} + \sin \frac{\varphi}{2} \cos \frac{\vartheta}{2} \cos \frac{\varphi}{2} \end{pmatrix}$$

$$= \begin{pmatrix} \cos \frac{\vartheta}{2} \cos \left(\frac{\varphi+\varphi}{2} \right) & \sin \frac{\vartheta}{2} \cos \left(\frac{\varphi+\varphi}{2} \right) & \sin \frac{\vartheta}{2} \sin \left(\frac{\varphi-\varphi}{2} \right) \\ \sin \frac{\vartheta}{2} \cos \left(\frac{\varphi-\varphi}{2} \right) & \cos \frac{\vartheta}{2} \sin \left(\frac{\varphi+\varphi}{2} \right) & \sin \frac{\vartheta}{2} \sin \left(\frac{\varphi-\varphi}{2} \right) \end{pmatrix} = \begin{pmatrix} \vartheta_0 \\ \vartheta_1 \\ \vartheta_2 \\ \vartheta_3 \end{pmatrix}$$

Угол конического поворота: $\alpha = 2 \arccos \vartheta_0$, $\vartheta_0 = \cos \frac{\alpha}{2}$, $\sin \frac{\alpha}{2} = \sqrt{1 - \vartheta_0^2}$

$$\Lambda_{\text{рез}} = \cos \frac{\alpha}{2} + \begin{pmatrix} j_1 \\ j_2 \\ j_3 \end{pmatrix} \sin \frac{\alpha}{2}; \quad j_1 \sqrt{1 - \vartheta_0^2} = \vartheta_1, \quad j_2 = \sqrt{1 - \vartheta_0^2} \vartheta_2, \quad j_3 = \sqrt{1 - \vartheta_0^2} \vartheta_3$$

Ответ: $j_1 = \frac{\vartheta_1^2}{\sqrt{1 - \vartheta_0^2}}$, $j_2 = \frac{\vartheta_2^2}{\sqrt{1 - \vartheta_0^2}}$, $j_3 = \frac{\vartheta_3^2}{\sqrt{1 - \vartheta_0^2}}$, $\alpha = 2 \arccos \vartheta_0$

где $\vartheta_0, \vartheta_1, \vartheta_2, \vartheta_3$ равны (см. выше).

N 4.70

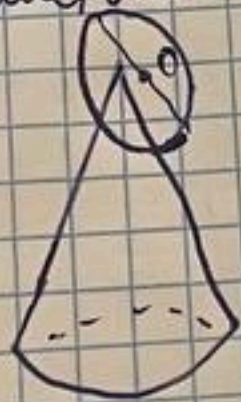
Дано:

$$\Lambda, \alpha = 90^\circ$$

$$t = \frac{\pi}{\Lambda}$$

$\vartheta_0, \vartheta_1, \vartheta_2, \vartheta_3$ - ?

Решение:



$$2m\vec{r}\vec{r}'\vec{\omega} = \vec{r} \times m\vec{g} + \vec{J}_0; \quad 2m\vec{r}'\vec{\omega} = m\vec{g}\vec{r}\sin\alpha; \quad 2\vec{r}'\vec{\omega} = g\sin\alpha = g\sin\omega t$$

$$2\vec{r}' = 2\omega\vec{r} = \cos\omega t \cdot \vec{\omega}; \quad r = \frac{\cos\omega t}{2\omega}; \quad 2L = \frac{\cos\omega t}{2\omega}; \quad 4L\omega^2 = \cos\omega t;$$

$$\omega t = \arccos 4L\omega^2; \quad t = \frac{\arccos 4L\omega^2}{\omega}; \quad \text{Омбери: } t = \frac{\arccos 4L\omega^2}{\omega}$$

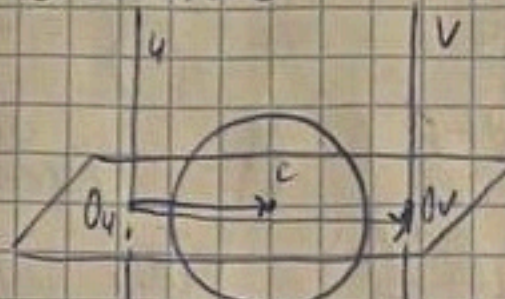
Дано:

$$J_v = J_u + m(\vec{r} \cdot \vec{r} - 2\vec{r} \cdot \vec{r}_c)$$

$$\vec{r} = \vec{O_u O_v}$$

$$\vec{r}_c = \vec{O_u C}$$

Решение



Теорема Тейлора-Уинчестера

$$J' = J_c + \int (\vec{r}' \cdot \vec{E} - \vec{r}' \times \vec{r}') dm$$

$$J_u = J_c + \int (\vec{O_u O_v} \cdot \vec{E} - \vec{O_u O_v} \times \vec{O_u O_v}) dm$$

$$J_v = J_c + \int (\vec{O_v O_v} \cdot \vec{E} - \vec{O_v O_v} \times \vec{O_v O_v}) dm$$

$$J_v = J_u + \int [(\vec{O_v O_v} - \vec{O_u O_v}) \cdot \vec{E} - \vec{O_v O_v} \times \vec{O_v O_v} + (\vec{O_u O_v} \times \vec{O_u O_v})] dm$$

$$\vec{O_v O_v} - \vec{O_u O_v} = (\vec{O_v} - \vec{O_u})(\vec{O_v} + \vec{O_u}) = (\vec{O_v} + \vec{O_u}) \cdot \vec{O_u O_v} =$$

$$= \vec{r}(\vec{O_u} + (\vec{O_u} + \vec{O_u O_v}) \cdot \vec{O_u O_v}) = \vec{r}(-\vec{r}_c - \vec{r}_c + \vec{r}) = \vec{r} \times \vec{r} -$$

$$- 2\vec{r}_c = -2\vec{r} \times \vec{r}_c = \vec{r} \times \vec{r} - 2\vec{r} \times \vec{r}_c$$

$$\vec{O_u} \times (\vec{O_u})^T - \vec{O_v} (\vec{O_v})^T = (\vec{O_u} - \vec{O_v})(\vec{O_u} + \vec{O_v}) =$$

$$= \begin{pmatrix} x_u \\ y_u \\ z_u \end{pmatrix} \cdot \begin{pmatrix} x_u & y_u & z_u \\ y_u & y_u & y_u \\ z_u & z_u & z_u \end{pmatrix} - \begin{pmatrix} x_v \\ y_v \\ z_v \end{pmatrix} \cdot \begin{pmatrix} x_v & y_v & z_v \\ y_v & y_v & y_v \\ z_v & z_v & z_v \end{pmatrix} = \begin{pmatrix} x_u^2 - x_v^2 & x_u y_u - x_v y_v & x_u z_u - x_v z_v \\ x_u y_u - x_v y_v & y_u^2 - y_v^2 & y_u z_u - y_v z_v \\ x_u z_u - x_v z_v & y_u z_u - y_v z_v & z_u^2 - z_v^2 \end{pmatrix}$$

$$\vec{r} = (x_v - x_u, y_v - y_u, z_v - z_u)$$

$$\vec{r}_c = (-x_u, -y_u, -z_u)$$

$$\vec{O_v}^2 - \vec{O_u}^2 = x_v^2 + y_v^2 + z_v^2 - x_u^2 - y_u^2 - z_u^2$$

$$= \begin{pmatrix} y_v^2 + z_v^2 - y_u^2 - z_u^2 & x_u y_u - x_v y_v & x_u z_u - x_v z_v \\ x_u y_u - x_v y_v & x_u^2 + z_u^2 - x_v^2 - z_v^2 & y_u z_u - y_v z_v \\ x_u z_u - x_v z_v & y_u z_u - y_v z_v & x_u^2 + y_u^2 - x_v^2 - y_v^2 \end{pmatrix}$$

П.К. O_u, C, O_v лежат в одной плоскости, тогда $z = 0$ ($z = 0$ $g \cdot l$)
 $z_u = 0 = z_v$

$$\vec{r} \cdot \vec{r} = \vec{r}^T \times \vec{r} = \begin{pmatrix} x_v - x_u & y_v - y_u & 0 \\ y_v - y_u & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \cdot \begin{pmatrix} x_v - x_u \\ y_v - y_u \\ 0 \end{pmatrix} = \begin{pmatrix} (x_v - x_u)^2 \\ (y_v - y_u)^2 \\ 0 \end{pmatrix}$$

Итог: $2\vec{r} \cdot \vec{r}_c = \begin{pmatrix} (x_u - x_v) x_u \\ (y_u - y_v) y_u \\ 0 \end{pmatrix}$

$$\vec{r} \cdot \vec{r} - 2\vec{r} \cdot \vec{r}_c = (x_v - x_u)^2 + (y_v - y_u)^2 - (x_u - x_v) \cdot 2x_u - (y_u - y_v) \cdot 2y_u =$$

$$= x_v^2 + x_u^2 - 2x_v x_u + y_v^2 + y_u^2 - 2y_v y_u + 2x_v x_u - 2x_u^2 + 2y_v y_u - 2y_u^2 =$$

$$= x_v^2 - x_u^2 + y_v^2 - y_u^2$$

$$\text{В итоге получаем: } \frac{x_u}{x_v} = \frac{y_u}{y_v} \rightarrow x_u$$

$$J_z \text{ мерзона инерции: } J_{zz} = \int (x_v^2 + y_v^2 - y_u^2 - x_u^2) dm$$

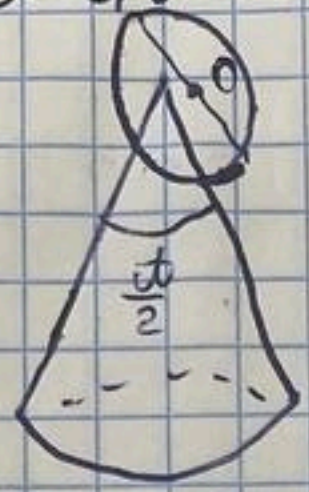
Dato:

$$\alpha, \alpha = 90^\circ$$

$$t = \frac{\pi}{\omega}$$

$\alpha_1, \alpha_2, \alpha_3 = ?$

Problema:



$$\alpha_1 = \cos \frac{\pi}{2} + \begin{pmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ 1/\sqrt{2} \end{pmatrix} \sin \frac{\pi}{2};$$

$$\alpha_2 = \cos \left(\frac{\pi}{\omega} \cdot \frac{1}{2\sqrt{2}} \right) + \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \sin \frac{\pi}{\sqrt{2}} =$$

$$\alpha_{\text{mez}} = \alpha_1 \alpha_2 = \begin{pmatrix} 0 \\ 1/\sqrt{2} \\ 0 \\ 1/\sqrt{2} \end{pmatrix} \begin{pmatrix} \cos \frac{\pi}{\sqrt{2}} \\ 0 \\ 0 \\ \sin \frac{\pi}{\sqrt{2}} \end{pmatrix} = \begin{pmatrix} 0 - \frac{1}{\sqrt{2}} \sin \frac{\pi}{\sqrt{2}} \\ 1/\sqrt{2} \cos \frac{\pi}{\sqrt{2}} \\ -1/\sqrt{2} \sin \frac{\pi}{\sqrt{2}} \\ 1/\sqrt{2} \cos \frac{\pi}{\sqrt{2}} \end{pmatrix}$$

$$\cos \alpha = -\cos^2 \frac{\pi}{8}$$

Answer: $\alpha = \arccos \left(\cos \left(-\frac{1}{\sqrt{2}} \sin \frac{\pi}{\sqrt{2}} \right) \right); f_1 = \frac{\frac{1}{2} \cos^2 \frac{\pi}{8}}{\sqrt{1 + \cos^2 \frac{\pi}{8}}} = f_3 \neq; f_2 = \frac{\sin \left(\frac{\pi}{\sqrt{2}} \right)}{\sqrt{1 + \cos^2 \frac{\pi}{8}}}$

Dato:

$$\alpha(t) = \cos \frac{\varphi(t)}{2} + e \sin \frac{\varphi(t)}{2}$$

$\omega = ?$

Problema:

N/4.85

$$\alpha(t) = \cos \frac{\varphi(t)}{2} + \bar{e} \sin \frac{\varphi(t)}{2}; \alpha^{-1} = \cos \frac{\varphi}{2} - \bar{e} \sin \frac{\varphi}{2}$$

$$\dot{\alpha}(t) = -\dot{\varphi}(t) \sin \frac{\varphi(t)}{2} \cdot \frac{1}{2} + \dot{\bar{e}} \sin \frac{\varphi(t)}{2} + \bar{e} \cos \frac{\varphi(t)}{2} \cdot \dot{\varphi}(t) \cdot \frac{1}{2}$$

$$\dot{\alpha}(t) = \frac{1}{2} \alpha(t) \omega(t) \quad \omega = 2 \dot{\alpha} \alpha^{-1} \dot{\alpha} = 2 \dot{\alpha}^{-1} \alpha \dot{\alpha}$$

$$\frac{1}{2} \omega \left(\cos \frac{\varphi(t)}{2} + \bar{e} \sin \frac{\varphi(t)}{2} \right) = \frac{1}{2} \dot{\varphi}(t) \sin \frac{\varphi(t)}{2} \left(\bar{e} \sin \frac{\varphi(t)}{2} + \bar{e} \cos \frac{\varphi(t)}{2} \right) + \frac{\varphi(t)}{2}$$

$$\omega = -\dot{\varphi} \sin \frac{\varphi}{2} \cos \frac{\varphi}{2} - 2(\bar{e}, \dot{\bar{e}}) \sin^2 \frac{\varphi}{2} + e^2 \cos \frac{\varphi}{2} \sin \frac{\varphi}{2} \dot{\varphi} +$$

$$2\bar{e} \sin^2 \frac{\varphi}{2} \dot{\varphi} + 2\dot{\bar{e}} \sin \frac{\varphi}{2} \cos \frac{\varphi}{2} + \bar{e} \cos^2 \frac{\varphi}{2} \dot{\varphi} - [\dot{\bar{e}}, \bar{e}] \sin^2 \frac{\varphi}{2} \cdot 2 =$$

$$= -\frac{\dot{\varphi}}{2} \sin \varphi + \dot{\bar{e}} \sin \varphi + \cancel{e^2 \frac{\dot{\varphi}}{2} \sin \varphi} + \bar{e} \sin^2 \frac{\varphi}{2} \dot{\varphi} + 2[\bar{e}, \dot{\bar{e}}] \sin^2 \frac{\varphi}{2} +$$

$$+ \bar{e} \dot{\varphi} \cos^2 \frac{\varphi}{2} = \dot{\bar{e}} \sin \varphi + [\bar{e}, \dot{\bar{e}}] (1 - \cos \varphi) + \bar{e} \cdot \dot{\varphi}$$

Answer: $\dot{\bar{e}} \sin \varphi + [\bar{e}, \dot{\bar{e}}] (1 - \cos \varphi) + \bar{e} \cdot \dot{\varphi}$

$$\frac{\partial F_x}{\partial y} \neq \frac{\partial F_y}{\partial x} \Rightarrow \text{не потенциальное.}$$

Ответ: не потенциальное.

Дано:

$$m, 2L$$

$$\omega$$

$$\omega = \text{const}$$

$$t = ?$$

$$\vec{v} = \vec{v}_0 + \vec{\omega} \times \vec{r}$$

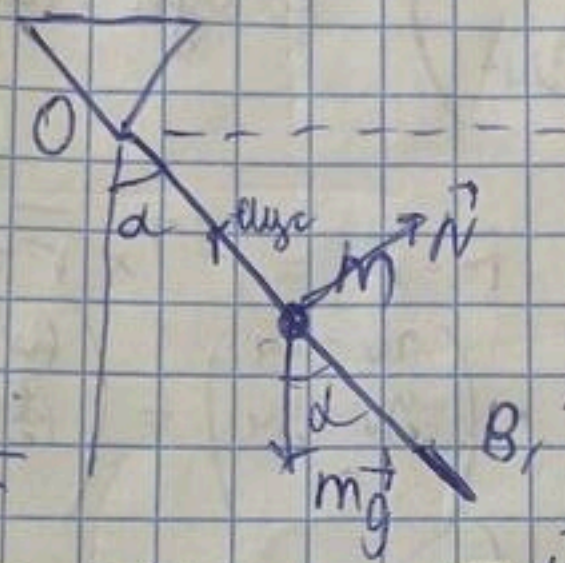
$$\frac{dK_0}{dt} = M_0; (mr^2 + J_0) \left(\frac{d\omega}{dt} \vec{e} + 0 \right) = \vec{r} \times m\vec{g} + J_0 \vec{e}$$

$$mr^2 \vec{e} + J_0 \vec{e} = \vec{r} \times m\vec{g} + J_0 \vec{e}; mr^2 \vec{e} = \vec{r} \times m\vec{g}; \vec{e} = \vec{r} \times \vec{g}$$

$$\vec{e} = \frac{1}{r^2} [\vec{r} \times \vec{g}]; \vec{\omega} = \frac{1}{r^2} [\vec{r} \times \vec{g}]$$

$$2r \vec{r}' \omega_m = \vec{r} \times \vec{g}; 2r' \omega_m = g \sin \alpha = g \sin \alpha \sin \alpha;$$

Решение:



$$J_0 = \frac{1}{3} m (2L)^2 = \frac{4}{3} mL^2$$

$$\vec{M}_0 = m\vec{g} \times \vec{r} = m\vec{g} \times \vec{r} + J_0 \vec{e}$$

$$K_0 = \vec{r} \times m\vec{v} + J_0 \cdot \vec{\omega} = (mr^2 + J_0) \omega$$

$$K_0 = mr^2 \omega + J_0 \omega = (mr^2 + J_0) \omega$$

$$2m\mathbf{r}\mathbf{r}'\dot{\omega} = \mathbf{r} \times m\mathbf{g} + \mathbf{I}\dot{\omega}; \quad 2m\mathbf{r}\mathbf{r}'\dot{\omega} = mgr\sin\alpha; \quad 2r'\dot{\omega} = g\sin\alpha = g\sin\omega t$$

$$2r' \rightarrow 2\omega r = \cos\omega t \cdot \frac{1}{\omega}; \quad r = \frac{\cos\omega t}{2\omega^2}; \quad 2L = \frac{\cos\omega t}{2\omega^2}; \quad 4L\omega^2 = \cos\omega t;$$

$$\omega t = \arccos 4L\omega^2; \quad t = \frac{\arccos 4L\omega^2}{\omega}; \quad \text{Orbitem: } t = \frac{4L\cos 4L\omega^2}{\omega}$$

N 11.1

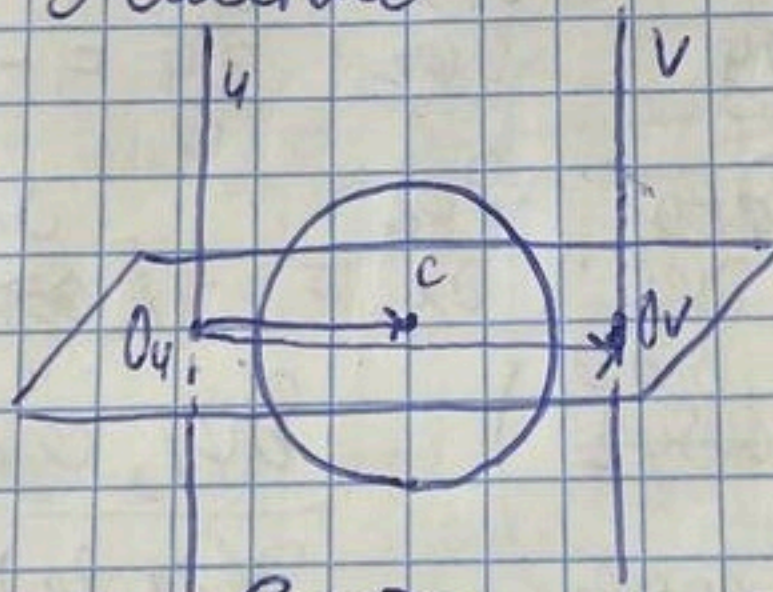
Dано:

$$J_v = J_u + m(\mathbf{r} \times \mathbf{r}' - 2\mathbf{r} \times \mathbf{r}_c)?$$

$$\mathbf{r} = \mathbf{O}_u \mathbf{O}_v$$

$$\mathbf{r}_c = \mathbf{O}_u \mathbf{C}$$

Решение



Теорема Штейнера-Гюльменера

$$J' = J_c + \int (\mathbf{r}^2 E - \mathbf{r} \times \mathbf{r}') dm$$

$$J_u = J_c + \int (\mathbf{O}_u^2 E - \mathbf{O}_u \times \mathbf{O}_u') dm$$

$$J_v = J_c + \int \mathbf{O}_v (\mathbf{O}_v^2 E - \mathbf{O}_v \times \mathbf{O}_v') dm$$

$$J_v = J_u + \int [(\mathbf{O}_v^2 - \mathbf{O}_u^2) E - \mathbf{O}_v \times \mathbf{O}_v' + \mathbf{O}_u \times \mathbf{O}_u'] dm$$

~~$$\mathbf{O}_v^2 - \mathbf{O}_u^2 = (\mathbf{O}_v - \mathbf{O}_u)(\mathbf{O}_v + \mathbf{O}_u) = (\mathbf{O}_v + \mathbf{O}_u) \cdot \mathbf{O}_u \mathbf{O}_v =$$

$$= \mathbf{r}(\mathbf{O}_u + \mathbf{O}_u + \mathbf{O}_u \mathbf{O}_v) \mathbf{O}_u \mathbf{O}_v = \mathbf{r}(-\mathbf{r}_c - \mathbf{r}_c + \mathbf{r}) = \mathbf{r} \times \mathbf{r} -$$

$$- 2\mathbf{r}_c - 2\mathbf{r} \times \mathbf{r}_c = \mathbf{r} \times \mathbf{r} - 2\mathbf{r} \times \mathbf{r}_c$$~~